Fake Promise-to-Pay Detection

Solution report for Problem Statement 1

Team Outlier Outlaws | CreditNirvana

1. Problem and approach

Collection teams record a large number of **Promises to Pay (PTPs)**. Many of them are fake: the borrower only wants to end the call, the agent pushed or entered a promise that was never made, or the borrower has a long history of promising and not paying. Our task is to score each PTP as it is captured so fake ones can be caught early.

Our approach followed one rule: do not build anything before it has been checked. The steps were:

- Build the label from real payments, and check it against the human annotations.
- Test each idea statistically (agent behaviour, language, finances, digital activity) and turn only the useful ones into features.
- Compare six models using a metric that suits an imbalanced problem.

All statistics, thresholds and encodings were fitted on the **training split** only. The validation split was used to choose the decision threshold, and the test split was used once at the end.

Data file	Rows	Used for
ptps	3,696	One row per PTP (the unit of analysis)
dial_attempts	51,105	Call details such as talk time and who hung up
call_transcripts	39,194 turns	All language features
accounts, lenders, agents, splits	2,400 / 6 / 30 / 2,400	Borrower finances, lender type, agent details, train/ validation/test split
payments	2,162	Building the label
post_call_events	8,466	Digital activity after the call
annotated_ptp_sample	200	Human labels, used to validate and explore

Our analysis ran in eight stages. The table below shows where each stage is covered in this report, and every figure produced by the analysis notebook is included (Figures 1 to 15).

Analysis stage	What it covers	Where in this report
Data Ingestion & Unification	Loading ten files, standardising dates, building one table, building the target	Sections 2.1 to 2.3
Target Validation & Portfolio Bias Assessment	Label balance and reliability, distributions, portfolio and lender bias, correlations	Sections 3, 3.3 and 4.1 to 4.5
Agent Behavior & Channel Dynamics	Bot or human, shift bunching, language team, agent tenure	Section 5 and 5.1
Live Conversational Linguistics (NLP)	Cleaning, code-mixing, word lists, date ownership, hang-up timing	Section 6 and 6.3
Financial History & Capacity to Pay	Historical keep rate, repeat promisers, salary proximity, hardship	Section 7 and 7.1
Post-Call Engagement Tracking	The 24-hour digital window and the intent index	Section 8 and 8.1
Final Feature Matrix Export	The columns passed to modelling	Section 9
End-to-End Machine Learning Pipeline	Preprocessing, tuning, six models, curves, saved pipeline	Section 10 and 10.4

Key results at a glance.

Result	Value
PTPs kept (payment-based label)	27.9% kept and 72.1% not kept
Human labels against real payments	Cramér's V = 0.56, chi-square p < 0.0001
Strongest language cue at capture	Firm commitment: 45.9% kept against 23.1% without it (+22.8 pp)
Strongest signal overall (after the call)	Intent index −1 to 2: fulfilment rises from 14.6% to 53.9%
Final model (XGBoost, test set)	PR-AUC 0.8341, F1 0.8458, precision 0.7425, recall 0.9826

2. Data preparation and the target

2.1 Loading and merging

We loaded the ten files, checked their shapes and standardised every timestamp. The files mixed ISO dates with day-first dates, so a single date parser would have misread values like 03/04. A small helper function (parse_ts) separates the two formats with a pattern match and parses each on its own.

We then joined dial attempts, accounts, lenders, agents, splits and annotations onto the PTP table with left joins, giving a table of 3,696 rows and 52 columns. Left joins keep every PTP even when some details are missing. Several tables share column names (town_id, dialling_arm, channel), so each join only brings in columns that are not already present.

2.2 Defining a kept promise

A PTP is **kept** if payments made between the capture time and three days after the promised date add up to at least **80% of the promised amount**. If the promised amount is missing we use the EMI amount. Under this rule **27.9%** of PTPs are kept and **72.1%** are not.

We used payments rather than agent remarks because payments are objective, while agent entries can themselves be wrong. The 80% and 3-day values are business assumptions: borrowers often pay slightly less or a day late, and a strict rule would call those promises fake.

2.3 Data Ingestion & Unification: detailed walkthrough

This part records, step by step, how the single analysis table was built, so that the work can be repeated.

Environment and helper functions. The analysis uses pandas and NumPy for the data, SciPy for the statistical tests, and seaborn and matplotlib for the figures. Two helpers are defined before anything else. **parse_ts** is the mixed-date parser described in 2.1. **wilson_ci** returns the 95% Wilson score interval for a proportion, which stays between 0% and 100% and is more reliable than the usual interval when a rate rests on few cases. It is used whenever a keep rate is reported.

Shape of each raw file. All ten files were loaded and their sizes checked before any merging.

File	Rows	Columns	Role
ptps	3,696	13	One row per PTP (the unit of analysis)
dial_attempts	51,105	16	Call details such as ring time, talk time and who hung up
call_transcripts	39,194	8	One row per spoken turn
accounts	2,400	20	Borrower finances and payment history
agents	30	6	Agent channel, language team, tenure and shift
lenders	6	4	Lender name and type
splits	2,400	2	Train, validation and test assignment per account
payments	2,162	5	Used to build the label
post_call_events	8,466	3	Digital activity after the call
annotated_ptp_sample	200	3	Human labels and annotator confidence

Timestamp standardisation. Five date columns were converted with parse_ts: captured_ts and promised_date (ptps), attempt_ts (dial_attempts), payment_ts (payments) and event_ts (post_call_events). Values that start with a four-digit year are read as ISO dates and everything else is read day-first. The first rows after conversion confirm the result.

Row	captured_ts	promised_date
0	2026-04-01 10:54:30	2026-04-03
1	2026-04-01 09:11:48	2026-04-08
2	2026-04-01 12:29:36	2026-04-03

Building the master table. The PTP table is the base, and each other table is added with a left join. Only columns that are not already present are brought in, which avoids duplicate column names.

Step	Table joined	Join key	Notes
1	ptps (base)	-	3,696 rows, one per PTP
2	dial_attempts	attempt_id	Call details; overlapping columns are skipped
3	accounts	account_id	Finances and history; dialling_arm is not repeated
4	lenders	lender_id	Lender name and lender type
5	agents	agent_id	channel is dropped because the PTP table already has it
6	splits	account_id	Train, validation or test
7	annotated_ptp_sample	ptp_id	Human type and confidence for 200 PTPs only

The result is a table of **3,696 rows and 52 columns**. Accounts and agents both carry a town_id column, so two versions (town_id_x and town_id_y) remain in the table; neither is used as a feature. The 52 columns come from these sources:

Source	Columns added	Examples
PTP table	13	ptp_id, account_id, channel, agent_id, captured_ts, promised_date, promised_amount
dial_attempts	11	phone_id, attempt_ts, ring_duration_s, talk_duration_s, hangup_by, disposition
accounts	18	portfolio, income_type, dpd_start, emi_amount, salary_credit_day, last_bounce_reason, prev_ptp_count
lenders	3	lender_name, lender_type, kyc_address_format
agents	4	language_team, town_id_y, tenure_months, shift
splits	1	split
annotations	2	annotated_type, annotator_confidence
Total	**52**	

Building the target, step by step.

- Every PTP is matched to all payments on the same account.
- Only payments made between the capture time and three days after the promised date are kept.
- These payments are summed into paid_amount, which is 0 when no payment falls in the window.
- The target amount is the promised amount, or the EMI amount when the promised amount is missing.
- is_kept is 1 when paid_amount is at least 80% of the target amount, and 0 otherwise. This gives 72.1% not kept and 27.9% kept.

Leakage guard. A training view (train_df) is created straight after the target is built. Every later statistic, test and threshold in the exploratory work uses this view only, so the validation and test targets stay unseen until the modelling stage.

3. Checking the label

Only 200 PTPs carry a human label (139 in train), so the annotations are used to understand and validate the data, not to train the model. The six label types in train are genuine_feasible (38), genuine_infeasible (34), escape (24), repeat_promiser (18), third_party (15) and agent_recorded_or_pushed (10).

3.1 Balance and label confidence

- **Shannon entropy** is 2.448 bits against a maximum of 2.585, giving a normalised value of 0.947 (about 5.5 effective classes out of 6). **Gini impurity** is 0.802. Both say the six types are fairly balanced, so no resampling is needed.
- A **chi-square test** between label type and annotator confidence gave p = 0.275 and **Cramér's V** of 0.21 (V = square root of chi-square divided by n times min(rows, columns) minus 1). Confidence does not depend on type, so label noise is not concentrated in one class.
- Only 9 of 2,572 train rows have low confidence. We still added a confidence weight (high 1.0, medium 0.7, low 0.3). Kish's effective sample size, (sum of w) squared over (sum of w squared), kept 93 to 97% of each class, so weighting loses very little.

3.2 Do human labels agree with real payments?

They agree strongly (chi-square p < 0.0001, Cramér's V = 0.56). Promises labelled escape or repeat_promiser were never kept, while genuine_feasible promises were kept 57.9% of the time. This supports using payments as the target.

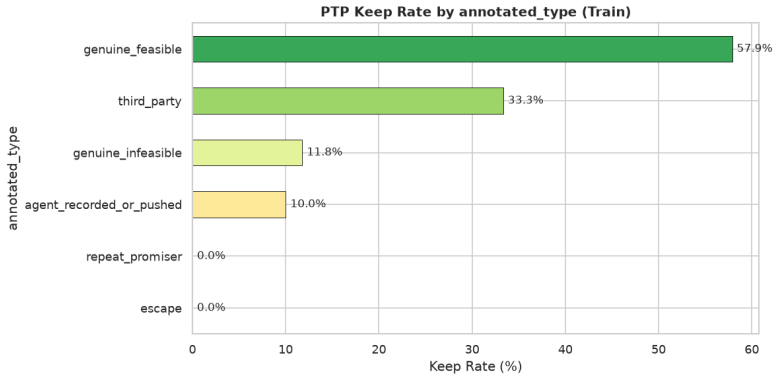


Figure 1. Keep rate by annotated PTP type (train)

3.3 Target Validation: class balance, annotator confidence and keep rates

Sections 3.1 and 3.2 give the headline results. The tables and figures below show the full numbers behind them. All of them use the 139 labelled PTPs in the training split.

Class frequency and imbalance ratio. The imbalance ratio is the size of the largest class divided by the size of each class.

Annotated type	Count	Share of labelled (%)	Imbalance ratio
genuine_feasible	38	27.34	1.00
genuine_infeasible	34	24.46	1.12
escape	24	17.27	1.58
repeat_promiser	18	12.95	2.11
third_party	15	10.79	2.53
agent_recorded_or_pushed	10	7.19	3.80

The largest class is genuine_feasible (38) and the smallest is agent_recorded_or_pushed (10), an overall ratio of 3.80 between them. Figure 2 shows the counts and shares.

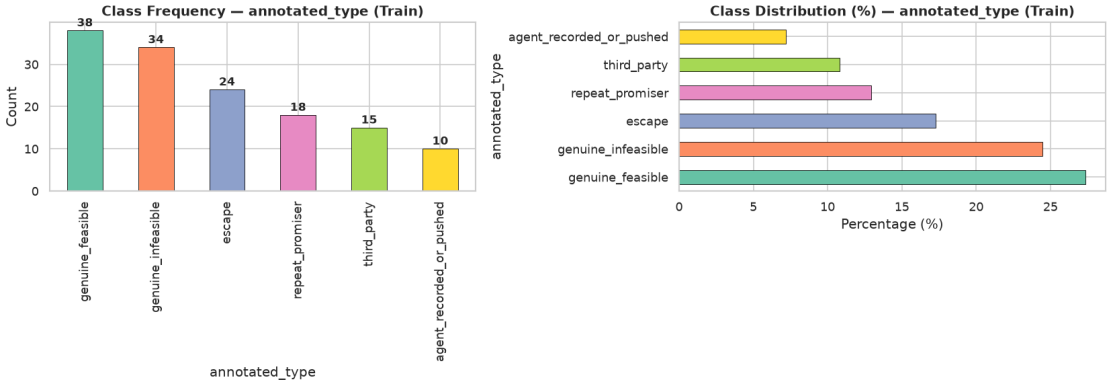


Figure 2. Class frequency and share of annotated PTP types (train)

Imbalance measures. Shannon entropy and Gini impurity put the same message into single numbers. A normalised entropy above 0.9 counts as fairly balanced, between 0.7 and 0.9 as mildly imbalanced, and below 0.7 as needing resampling or class weights.

Measure	Value	Reading
Number of classes (k)	6	
Shannon entropy H	2.4476 bits	
Maximum entropy (all classes equal)	2.5850 bits	
Normalised entropy (H / maximum)	0.9469	Fairly balanced
Effective number of classes (2 to the power H)	5.46 of 6	
Gini impurity	0.8020	Fairly balanced

Annotator confidence by PTP type. The table counts how often each type was labelled with high, medium or low confidence.

Annotated type	High	Medium	Low	Total
agent_recorded_or_pushed	5	5	0	10
escape	14	8	2	24
genuine_feasible	24	12	2	38
genuine_infeasible	15	16	3	34
repeat_promiser	6	11	1	18
third_party	12	2	1	15
Total	**76**	**54**	**9**	**139**

The chi-square statistic is 12.1515 with p = 0.2750 and Cramér's V = 0.2091, so confidence does not depend on type. Figure 3 shows the same table as counts and as row percentages. The third_party type has the highest share of high-confidence labels (80.0%) and repeat_promiser the lowest (33.3%), but none of these gaps is statistically significant.

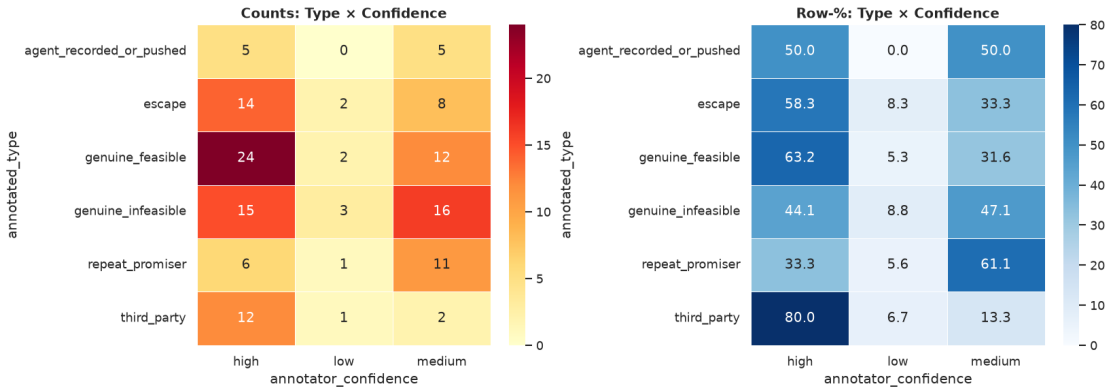


Figure 3. Annotator confidence by PTP type, counts and row percentages (train)

Low-confidence labels. Entries with low confidence are flagged in a column called is_unreliable_label. There are 9 of them among 2,572 training rows (0.3%). The share of low-confidence labels within each type is small everywhere.

Annotated type	Share labelled low confidence (%)
genuine_infeasible	8.8
escape	8.3
third_party	6.7
repeat_promiser	5.6
genuine_feasible	5.3
agent_recorded_or_pushed	0.0

Confidence weights and effective sample size. High, medium and low confidence map to weights of 1.0, 0.7 and 0.3 (PTPs with no label get 1.0). Kish's effective sample size shows how much information remains after weighting.

Annotated type	Raw n	Effective n (Kish)	Retained (%)
agent_recorded_or_pushed	10	9.7	97.0
escape	24	22.5	93.9
genuine_feasible	38	36.2	95.3
genuine_infeasible	34	31.8	93.5
repeat_promiser	18	17.1	94.9
third_party	15	14.4	95.7

Lender-type encoding. Lender type is one-hot encoded into six columns (Fintech NBFC, NBFC, NBFC (MSME), NBFC-MFI, Private bank and Small finance bank). A check confirmed that every row has exactly one of them set. They are kept as cheap controls.

The numbers behind Figure 1. Each row shows how often PTPs of that annotated type were kept or not kept in reality.

Annotated type	Not kept (%)	Kept (%)
agent_recorded_or_pushed	90.0	10.0
escape	100.0	0.0
genuine_feasible	42.1	57.9
genuine_infeasible	88.2	11.8
repeat_promiser	100.0	0.0
third_party	66.7	33.3

The chi-square statistic is 42.9260 with $p < 0.0001$ and Cramér's V = 0.5557.

4. Exploratory analysis

- **Distributions.** Talk time, ring time and selection propensity all failed the Shapiro-Wilk normality test, and selection propensity is almost constant at 1.0 (skewness -5.3). This is why we used rank-based tests and tree models.
- **Kruskal-Wallis test** (a rank-based alternative to ANOVA, effect size eta squared = (H - k + 1) / (n - k)). Talk time differs across label types (H = 37.1, $p < 0.0001$, eta squared 0.24). Ring time ($p = 0.095$) and selection propensity ($p = 0.725$) do not, so they were dropped.
- **Bias checks.** Chi-square tests against label type found no significant effect for portfolio ($p = 0.23$), lender type ($p = 0.57$), channel ($p = 0.58$) or hang-up party ($p = 0.75$). Only income type was significant ($p = 0.023$).
- **Correlations.** The Spearman matrix (Figure 4) shows the money columns move together (promised amount and overdue amount, 0.96), so we avoided raw amounts. It also shows paid_amount correlates 0.81 with is_kept, which is true by construction, so it was kept out of the model to avoid leakage.

A caution: with 139 labelled rows and cross-tabs of up to 36 cells, many expected counts are small. These tests point in a direction but are not conclusive.

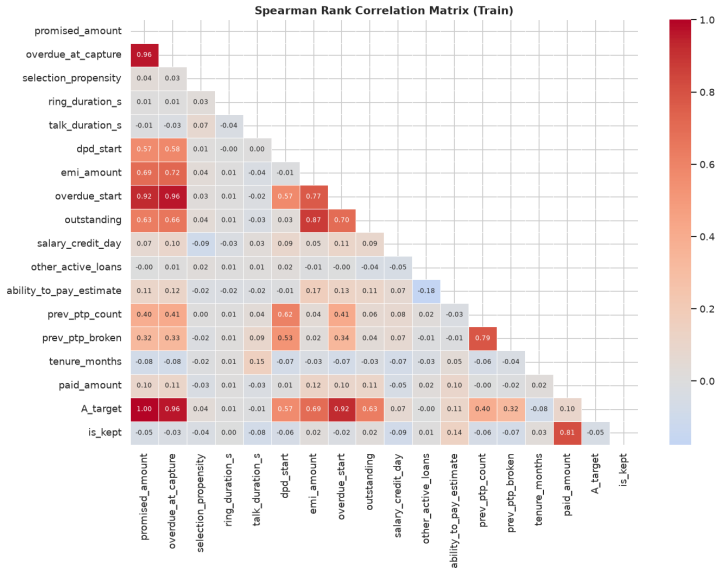


Figure 4. Spearman correlation matrix (train)

4.1 Univariate analysis of continuous variables

The statistics below describe the call variables in the training split (2,147 calls with a dial record). They show skewness, heavy tails and spikes, which decide whether rank-based tests and tree models are needed.

Variable	n	Mean	Std	Min	25%	Median	75%	Max
talk_duration_s	2,147	43.56	13.80	0	33	43	53	91
ring_duration_s	2,147	11.76	5.52	2	7	12	16	43
selection_propensity	2,147	0.98	0.11	0.25	1	1	1	1

Variable	Skewness	Kurtosis	IQR	CV (%)	Shapiro-Wilk W	Result
talk_duration_s	0.108	0.170	20	31.68	0.9904	Not normal
ring_duration_s	0.473	0.906	9	46.94	0.9450	Not normal
selection_propensity	−5.287	26.805	0	10.82	0.1797	Not normal

All three variables fail the Shapiro-Wilk test ($p < 0.0001$). Talk time is roughly bell-shaped with a small spike at 0 seconds. Ring time spreads from 2 to 43 seconds with two humps. Selection propensity sits at 1.0 for almost every call, with a few low values (minimum 0.25). Figure 5 shows each distribution with its density curve, mean and median.

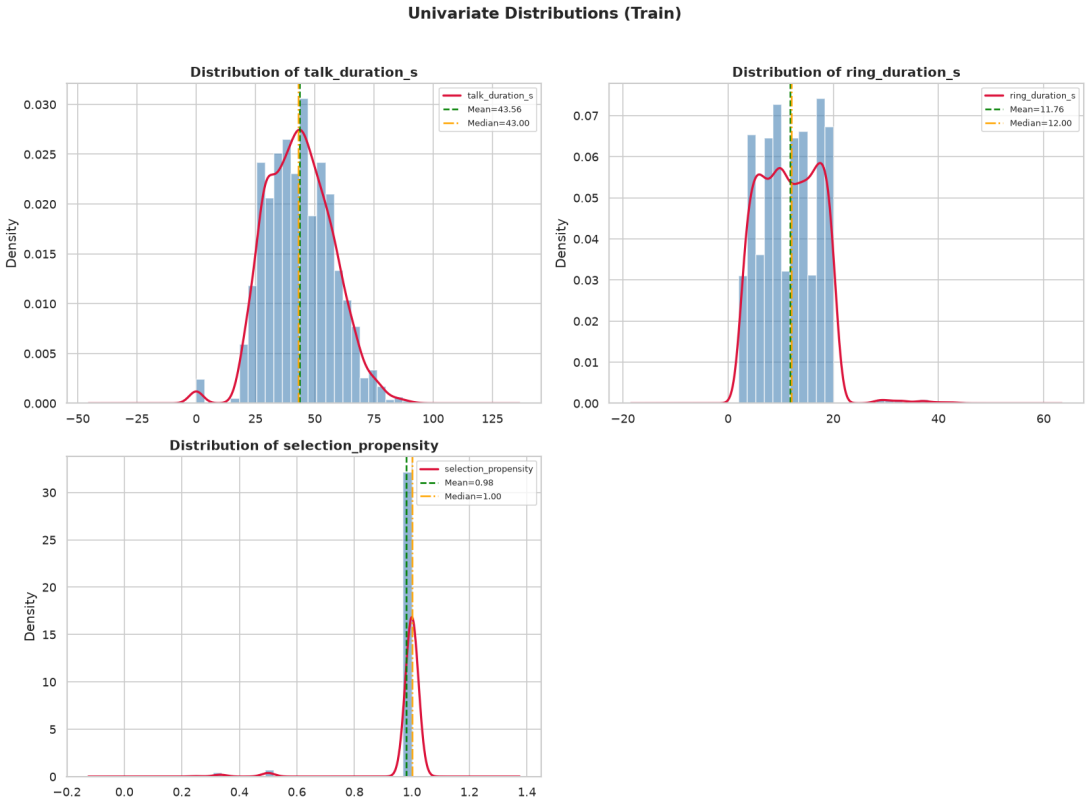


Figure 5. Distributions of talk time, ring time and selection propensity with density curves (train)

4.2 Categorical profile

Category counts and shares in the training split. Every category is populated, and the smallest, NBFC (MSME), still has 125 rows.

Variable	Categories (count, share of rows)
portfolio (2,572)	pl_salaried 721 (28.0%); two_wheeler 514 (20.0%); mfi_jlg 431 (16.8%); consumer_durable 319 (12.4%); credit_card 304 (11.8%); msme 283 (11.0%)
income_type (2,572)	salaried 1,339 (52.1%); self_employed 856 (33.3%); daily_wage 377 (14.7%)
lender_type (2,572)	Private bank 661 (25.7%); Small finance bank 652 (25.3%); Fintech NBFC 527 (20.5%); NBFC 408 (15.9%); NBFC-MFI 199 (7.7%); NBFC (MSME) 125 (4.9%)
channel (2,572)	tele_agent 1,960 (76.2%); field 425 (16.5%); voice_bot 187 (7.3%)
hangup_by (2,147)	agent 1,175 (54.7%); customer 955 (44.5%); network 17 (0.8%)
disposition (2,147)	rpc_ptp 2,042 (95.1%); third_party_ptp 105 (4.9%)

4.3 Kruskal-Wallis test and violin plots

The Kruskal-Wallis test asks whether a continuous variable differs across the six label types without assuming a normal distribution. Only talk time separates the types.

Variable	H statistic	p-value	Eta squared	Significant
talk_duration_s	37.1390	< 0.0001	0.2416	Yes
ring_duration_s	9.3747	0.0950	0.0329	No
selection_propensity	2.8386	0.7249	−0.0163	No

The negative eta squared for selection propensity means the effect is effectively zero. Figure 6 shows the full spread for each type. Third-party calls are the shortest and the most tightly grouped. Selection propensity is flat at 1.0 for most types, which is why it carries no signal.

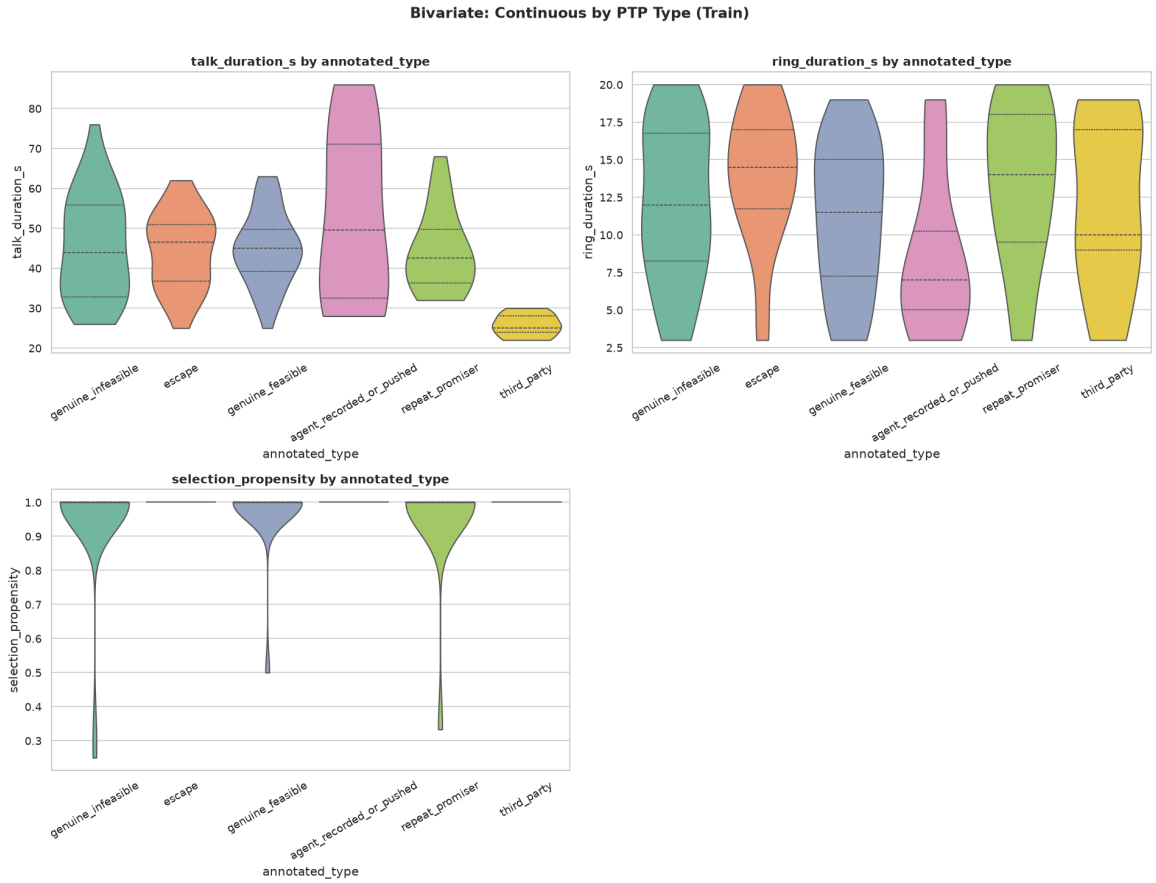


Figure 6. Talk time, ring time and selection propensity by annotated PTP type (train)

4.4 Portfolio Bias Assessment and lender-type bias

Some portfolios or lenders could attract more fake promises because of the borrowers they serve. If so, the model might learn the portfolio instead of the behaviour. We tested annotated type against each, using a chi-square test of independence.

Annotated type by portfolio (counts).

Annotated type	Consumer durable	Credit card	MFI-JLG	MSME	PL salaried	Two-wheeler
agent_recorded_or_pushed	3	1	2	2	2	0
escape	6	4	1	2	7	4
genuine_feasible	3	3	5	6	17	4
genuine_infeasible	5	3	8	5	8	5
repeat_promiser	2	1	5	4	3	3
third_party	1	4	6	2	1	1

Chi-square = 29.8533 with 25 degrees of freedom, p = 0.2298 and Cramér's V = 0.2073, so there is **no significant portfolio bias**. Figure 7 shows each portfolio as column percentages. Some gaps look large, for example 44.7% of PL salaried labels are genuine_feasible and 30.0% of consumer durable labels are escape, but each portfolio has only 16 to 38 labelled PTPs, so these gaps are not statistically reliable.

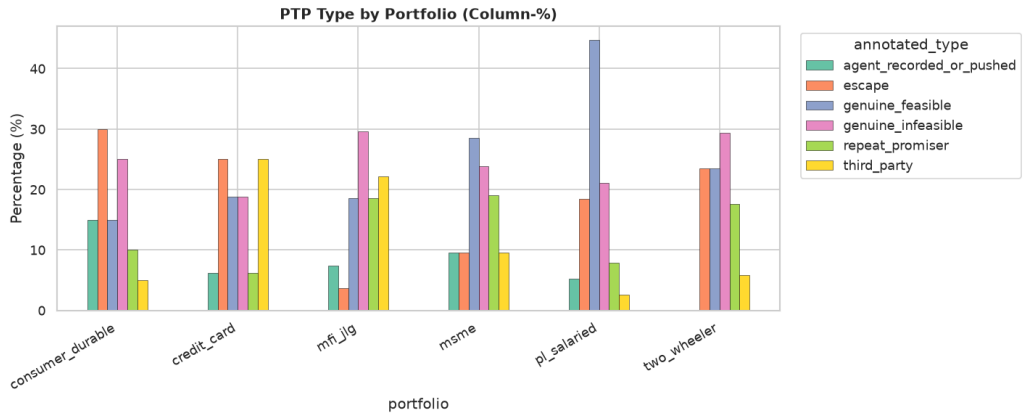


Figure 7. Annotated PTP type by portfolio, column percentages (train)

Annotated type by lender type (counts).

Annotated type	Fintech NBFC	NBFC	NBFC (MSME)	NBFC-MFI	Private bank	Small finance bank
agent_recorded_or_pushed	1	2	1	1	3	2
escape	8	5	0	1	7	3
genuine_feasible	11	3	2	1	11	10
genuine_infeasible	7	3	3	5	8	8
repeat_promiser	4	2	1	3	2	6
third_party	0	2	0	3	5	5

Chi-square = 23.1683 with 25 degrees of freedom, p = 0.5677 and Cramér's V = 0.1826, so there is **no significant lender-type bias** either. The highest cells in Figure 8 rest on very few PTPs (NBFC (MSME) has only 7 labelled PTPs, so its 42.9% genuine_infeasible share is 3 cases), and should not be read as real differences.

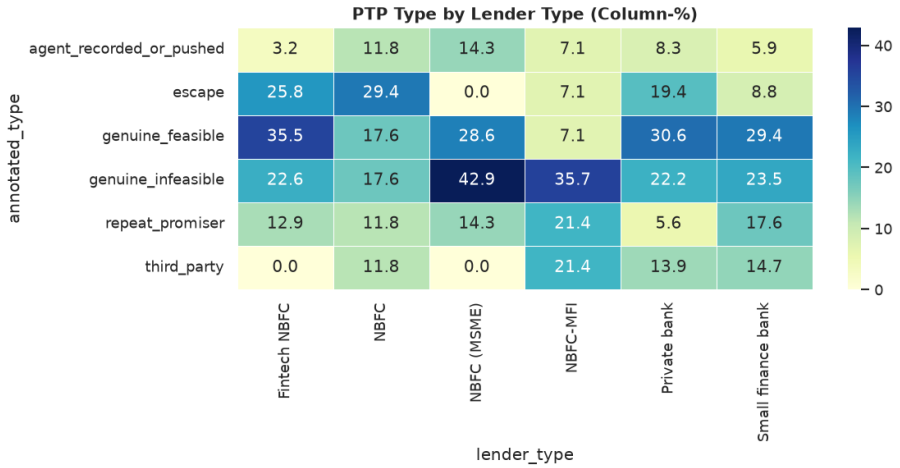


Figure 8. Annotated PTP type by lender type, column percentages (train)

4.5 Chi-square summary across the categorical variables

All categorical variables were tested against annotated type in one pass. Only income type is significant. The earlier tests cover portfolio, lender type, channel and hang-up party; four further columns named in the notebook list (ptp_initiative, is_conditional, discrepancy_flag and log_integrity) are not in the merged table and were skipped automatically.

Variable	Chi-square	dof	p-value	Cramér's V	Significant
portfolio	29.85	25	0.2298	0.2073	No
income_type	20.76	10	0.0228	0.2733	Yes
lender_type	23.17	25	0.5677	0.1826	No
channel	3.82	5	0.5752	0.1658	No
hangup_by	2.68	5	0.7490	0.1389	No

Because the labelled sample is small and the cross-tabs have many cells, a non-significant result here means "no evidence of bias", not proof that bias is absent.

5. Agent and channel features

- **Voice bot flag.** Bots (250 PTPs) behave differently from tele-agents (2,841) and field staff (605), so a flag lets the model separate them.
- **Shift bunching.** We guessed that agents rush near the end of a shift. The shift end was inferred as 19:00 from the data, and the last 30 minutes are flagged (80 of 3,446 human PTPs). Capture rate in that window is 2.67 per minute against 2.11 in the previous two hours, which supports the idea.
- **Language team.** Keep rates differ slightly by team (0.26 for English to 0.31 for the combined team), so each team's train keep rate is used as a feature, with the overall mean for unseen teams.
- **Agent tenure.** We compared fake rates by tenure with Wilson confidence intervals (used because each agent has few PTPs). The relationship was weak and not significant (Spearman -0.28, p = 0.23), so tenure was not used.

5.1 Agent Behavior & Channel Dynamics: detailed results

Channel split. A flag called is_bot marks voice-bot PTPs, so the model can separate them from human agents.

Channel	PTPs	Share of all 3,696 PTPs
tele_agent	2,841	76.9%
field	605	16.4%
voice_bot	250	6.8%

Shift bunching. The shift end is the 99.9th percentile of human capture times, rounded up to the next whole hour. PTPs captured in the last 30 minutes before that time are flagged.

Measure	Value
Inferred shift end	19:00
Flagged window	Last 30 minutes (from 18:30)
PTPs captured per minute, last 30 minutes	2.67
PTPs captured per minute, previous 2 hours	2.11
Human PTPs flagged	80 of 3,446

The capture rate in the last half hour is about 1.3 times the earlier rate, which supports the idea that agents rush near the end of a shift.

Language team encoding. Each team is replaced by its keep rate in the training split. A team not seen in training receives the overall training mean.

Language team	Keep rate (train)
all (combined team)	0.305
english	0.262
hinglish	0.284
kanglish	0.299

Agent tenure. For this test a PTP is counted as flagged when its annotated type is escape or agent_recorded_or_pushed. Only human-handled, labelled training PTPs are used, which gives 20 agents with 2 to 11 labelled PTPs each. The overall flagged rate is 24.2%.

Test	Result
Spearman correlation, tenure against fake rate (per agent)	ρ = −0.28, p = 0.228
Chi-square, tenure bands (up to 12 months, 13 to 36 months, over 36 months)	p = 0.351
agent_recorded_or_pushed PTPs in this sample	10

Figure 9 plots each agent's rate with its 95% Wilson interval. The intervals are wide, most overlap the overall rate, and there is no visible trend with tenure, so tenure was not used as a feature.

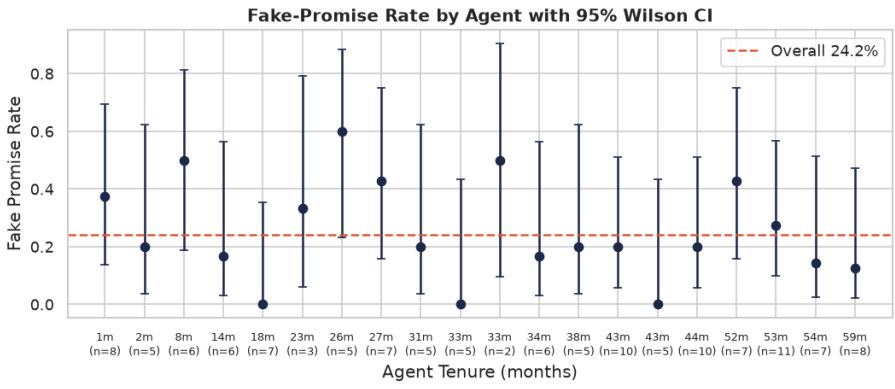


Figure 9. Fake-promise rate by agent with 95% Wilson interval, ordered by tenure

6. Language features from transcripts

6.1 Cleaning

Transcripts mix English, Hinglish and Kannada-English, and some text is in Devanagari. We lower-cased the text, converted common Devanagari phrases and digits to Roman script, removed tags and punctuation, and labelled each turn as agent or borrower. About 12.6% of borrower turns are code-mixed, so every word list includes Hindi and Kannada phrases as well as English.

6.2 Signals and keep rate

We built four word lists (hedging, conditional, vague timing, firm commitment) and compared keep rates when each was present or absent, using chi-square tests and Wilson intervals.

Signal	Share of PTPs	Keep rate if present	If absent	p-value
Firm commitment	23.4%	45.9%	23.1%	< 0.0001
Vague timing	29.5%	23.4%	30.5%	0.0003
Hedging	49.3%	26.3%	30.4%	0.024
Conditional	6.2%	23.8%	28.7%	0.207

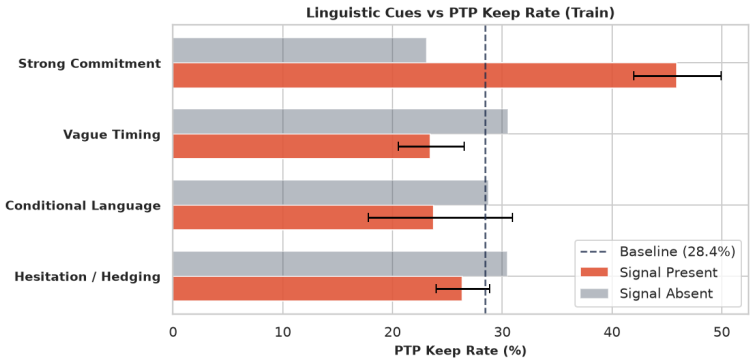


Figure 10. Keep rate with and without each language cue

- Pure uncertainty phrases such as "dekhta hoon" are the damaging ones (17.5% kept, against 29.5% for neutral calls). Conditional phrases like "salary aayi toh" are close to normal (27.9%).
- We expected promises where the borrower named the date to be stronger. The opposite was true (24.0% against 30.6% when the agent named it). We kept the feature and let the model decide rather than force our assumption.
- Time between agreement and hang-up was not significantly different (Mann-Whitney p = 0.106; medians 3.0 s for kept and 3.6 s for broken), so it is only a weak feature.

6.3 Live Conversational Linguistics (NLP): detailed walkthrough

Text normalisation. Every transcript turn goes through the same steps:

- 17 common Devanagari phrases (for example the ones for "dekhta hoon", "shayad" and "pakka") are replaced with Roman-script equivalents, and Devanagari digits become 0 to 9.
- Text is lower-cased, and tags such as [inaudible], [name] and [lender] are removed.
- Currency markers (₹, rs, inr) and punctuation are removed, letters repeated three or more times are cut to two, and extra spaces are collapsed.

Speaker labels. Tags such as agent, bot, executive and caller become AGENT, and every other speaker becomes BORROWER. After this step the data holds **39,194 turns across 3,367 calls**, sorted by call and turn. The first turns of one call show the result (text shortened).

Speaker	Normalised text
AGENT	hello collections team se bol rahi hoon ji bol...
BORROWER	haan boliye
AGENT	aapke loan account mein 6 000 rupaye overdue h...
AGENT	pichli baar bhi aapne 29 january ka bola tha p...
BORROWER	woh baat thi is baar ho jayega

Code-mixing audit. A borrower turn counts as code-mixed when it holds both Hindi and English words, or both Kannada and English words (small word lists of 20 Hindi, 19 English and 10 Kannada tokens). Only borrower turns linked to training PTPs were checked.

Measure	Value
Borrower turns in train	10,927
Code-mixed turns (Hinglish or Kanglish)	1,379
Share of borrower turns	12.62%

About one borrower turn in eight mixes languages, so every word list below contains Hindi and Kannada phrases as well as English.

The four word lists. A PTP gets a flag when any phrase of the list appears anywhere in the borrower's turns of that call.

List	Phrases	Example phrases
Hedging	23	dekhta hoon, koshish karunga, pata nahi, shayad, let me see, will try, maybe, nodtini
Conditional	13	salary aayi toh, paisa aaya toh, agar, jaise hi, bonus aaya toh, once salary
Vague timing	17	next week, agle hafte, month end, jaldi, soon, later, baad mein, mundina vara
Firm commitment	16	pakka, guarantee, definitely, zaroor, aaj hi, upi se kar dunga, pakka kodtini

Signal strength with confidence intervals. The table extends the one in Section 6.2 with counts, 95% Wilson intervals and the gap in percentage points. The base is the 2,572 training PTPs.

Signal	n	Share	Kept if present	95% interval	Kept if absent	Gap (pp)	p-value
Hedging	1,268	49.3%	26.3%	[24.0%, 28.8%]	30.4%	−4.1	0.0236
Conditional	160	6.2%	23.8%	[17.8%, 30.9%]	28.7%	−5.0	0.2068
Vague timing	760	29.5%	23.4%	[20.5%, 26.6%]	30.5%	−7.1	0.0003
Firm commitment	601	23.4%	45.9%	[42.0%, 49.9%]	23.1%	+22.8	< 0.0001

Uncertainty against conditional language. Hedging mixes two different things, so it was split. Category A is pure uncertainty (dekhta hoon, dekhti hoon, sochta hoon, dekhna padega, let me see, pata nahi), and it also takes any PTP that has both kinds of phrase. Category B is a specific condition (salary aayi toh, salary aane par, bonus aaya toh, paisa aaya toh, payment aane ke baad, once salary) without uncertainty. Category C has neither.

Category	n	Keep rate	95% interval
A: Uncertainty ("dekhta hoon")	228	17.5%	[13.2%, 23.0%]
B: Conditional ("salary aayi toh")	61	27.9%	[18.2%, 40.2%]
C: Neutral baseline	2,283	29.5%	[27.7%, 31.4%]

Uncertainty clearly lowers the keep rate, while conditional promises are close to the baseline. The interval for category B is wide because it has only 61 PTPs.

Who proposed the date. A regular expression finds the first explicit date in the call (for example "12th jan", "kal", "parso", "tomorrow", a weekday, "next week" or "month end"), and the speaker of that turn is the proposer. Of the 2,572 training PTPs, 1,042 had no clear date mention.

Date proposed by	n	Keep rate	95% interval
Borrower	313	24.0%	[19.6%, 29.0%]
Agent	1,217	30.6%	[28.1%, 33.3%]
Difference (borrower minus agent)		−6.7 pp	

Conversational abruptness (seconds_to_hangup). This is the time from the borrower's last agreement word (haan, ji, theek hai, thik, ok, okay, fine, kar dunga, pakka) to the end of the call. If there is no agreement word, the last borrower turn is used, and the value is never below 0. It could be computed for 3,067 of 3,696 PTPs. The idea was that a borrower who hangs up straight after agreeing only wants to end the call.

PTP outcome	Median	Middle half (IQR)	Test
Kept	3.0 s	0.0 to 8.1 s	Mann-Whitney U = 1,011,474
Broken	3.6 s	0.0 to 8.7 s	p = 0.1064

Broken promises show a slightly longer wait before hang-up, which is the opposite of the idea, and the difference is not significant (Figure 11). This is why it is only a weak feature.

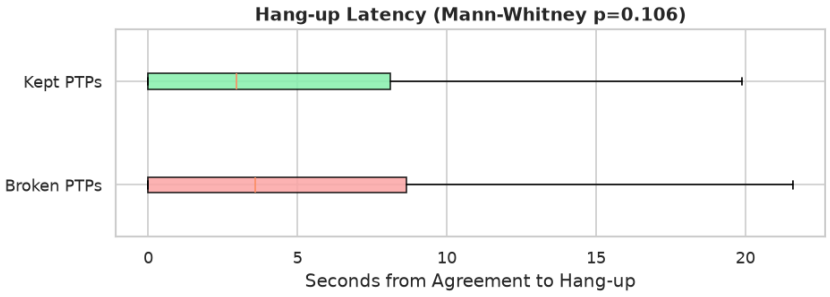


Figure 11. Hang-up latency for kept and broken PTPs (Mann-Whitney p = 0.106)

NLP features added to the master table. The features were built for every PTP in all three splits, not only training. The word-list flags are 1 or 0 per PTP:

Feature	PTPs flagged (all 3,696)
has_hedge	1,836
has_condition	234
has_vague	1,072
has_firm	877
category_A_uncertainty, category_B_conditional	Mutually exclusive flags
borrower_proposed_date, agent_proposed_date	Flags from the date-ownership step
seconds_to_hangup	3,067 non-missing values

7. Financial history and capacity to pay

- **Historical keep rate** is (past PTPs minus broken PTPs) divided by past PTPs, and 0 when there is no history. Past behaviour is usually the best predictor. Figure 12 shows three groups: accounts near 0, around 0.55, and near 1.
- **High-risk repeat promiser.** High-confidence repeat promisers have a median historical keep rate of 0, so accounts with at least 2 past PTPs and none kept are flagged (297 PTPs).
- **Salary proximity** is the distance between the promised day and the usual salary day, wrapped around the month: min(d, 30 - d), so day 29 and day 1 are 2 days apart. Feasible promises sit a median 2 days from payday, infeasible ones about 9 (Figure 13). Only 1,162 PTPs have a salary day, so missing values are filled with the median.
- **Hardship index** is a hardship score from the last bounce reason (2 for insufficient funds, 0 for technical issues, 1 otherwise) times log(1 + days past due). It varies little across label types, so it is a weak signal.

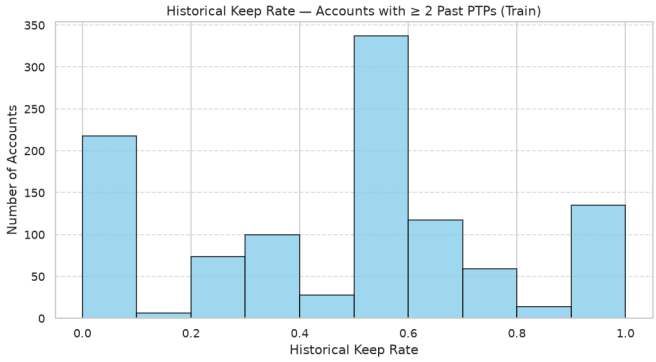


Figure 12. Historical keep rate, accounts with 2 or more past PTPs

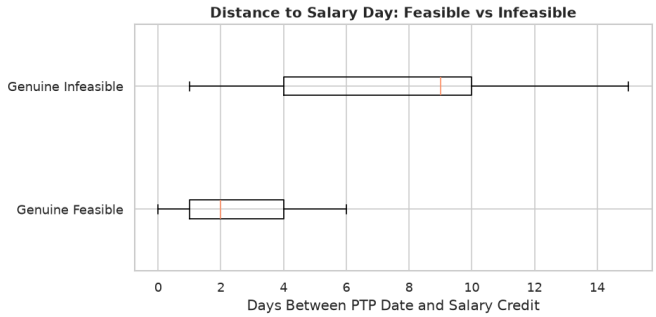


Figure 13. Distance from salary day, feasible vs infeasible promises

7.1 Financial History & Capacity to Pay: detailed results

Historical keep rate. There are 1,088 training PTPs whose accounts have at least 2 past PTPs; these are the PTPs shown in Figure 12. Where the history is missing, past PTP and broken counts are treated as 0.

Choosing the repeat-promiser rule. The rule was grounded in the human labels. Among PTPs labelled repeat_promiser, the median historical keep rate is zero at high and medium confidence. The single low-confidence case is the exception.

Annotator confidence (repeat_promiser)	Median historical keep rate
High	0.00
Medium	0.00
Low (1 PTP)	0.67

This gives the rule: at least 2 past PTPs and a historical keep rate of exactly 0. It flags **297 PTPs** across the data. Among the 200 labelled PTPs, 13 carry the flag.

Annotated type	Not flagged	Flagged	Total
agent_recorded_or_pushed	12	1	13
escape	35	0	35
genuine_feasible	53	1	54
genuine_infeasible	46	5	51
repeat_promiser	25	4	29
third_party	16	2	18
All	**187**	**13**	**200**

The flag overlaps only partly with the human labels, since it reflects payment history while labels reflect what was said on the call. It is therefore judged by how the model uses it.

Salary proximity. The score is computed only where both the usual salary day and the promised day are known. That is the case for 1,162 of 3,696 PTPs (31.4%), and the model fills the rest with the median. Figure 13 compares feasible and infeasible promises.

Hardship index. The hardship score comes from the text of the last bounce reason, and the index multiplies it by the natural log of 1 plus days past due (negative or missing days count as 0).

Last bounce reason contains	Hardship score
insufficient, funds, balance or debit	2
technical, network or system	0
Missing reason	0
Anything else	1

The mean index barely differs between annotated types, from 4.63 for third_party to 6.21 for agent_recorded_or_pushed, so it is a weak signal.

Annotated type	Mean composite hardship index
agent_recorded_or_pushed	6.21
escape	5.63
genuine_feasible	5.74
genuine_infeasible	5.43
repeat_promiser	6.10
third_party	4.63

8. Activity after the call

Of 8,466 post-call events, 5,541 happen within 24 hours of capture. We counted the types of activity per PTP and built an intent index: payment link opened + YES reply + inbound call − NO reply. Fulfilment rises steadily with the index.

Intent index	−1	0	1	2
PTPs	41	2,549	1,003	102
Fulfilment rate	14.6%	22.5%	39.5%	53.9%

The chi-square test gives p ≈ 1e-29. Opening the payment link alone raises fulfilment from 23.4% to 42.7%. This is the strongest behavioural signal we found, but it only exists after the call, so using it in the capture-time model would leak future information. We treat it as a second-stage check that re-scores a PTP after 24 hours.

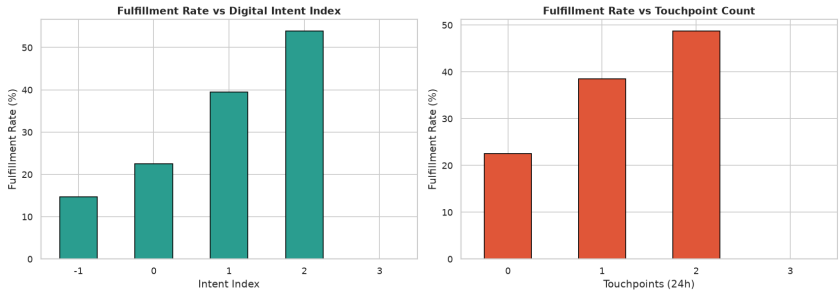


Figure 14. Fulfilment rate by intent index and touchpoint count

8.1 Post-Call Engagement Tracking: detailed results

The 24-hour window. Each post-call event is linked to its PTP, and only events from 0 to 24 hours after capture are used.

Measure	Value
Post-call events in total	8,466
Events within 24 hours of capture	5,541
PTPs with at least one event in 24 hours	2,944 (79.7% of 3,696)

Events in the window. Only the last four event types are counted in the intent index. The first two are messages we sent, so they say nothing about the borrower.

Event type	Events	Role in the intent index
payment_link_sent	2,393	Not counted (sent by us)
whatsapp_confirmation_sent	1,822	Not counted (sent by us)
payment_link_opened	900	+1 if present
confirmation_reply_yes	289	+1 if present
inbound_call_from_customer	79	+1 if present
confirmation_reply_no	58	−1 if present

Each of the four event types is counted once per PTP. The **digital intent index** is link opened + YES reply + inbound call − NO reply. The **touchpoint count** is how many of those four types occurred (0 to 4). Event timing is also recorded for every PTP: first event time, last event time, number of event types and total events. These are in the master table but are not part of the exported feature matrix.

Fulfilment by intent index (complete). The table in Section 8 shows indices −1 to 2. The single PTP with index 3 is added here for completeness.

Intent index	PTPs	Fulfilled	Fulfilment rate
−1	41	6	14.6%
0	2,549	574	22.5%
1	1,003	396	39.5%
2	102	55	53.9%
3	1	0	0.0%

The chi-square statistic is 141.90 with p = 1.1e-29.

Single touchpoints and total touchpoints. Each key signal was also tested on its own.

Signal	Group	PTPs	Fulfilled	Rate
Payment link opened	No	2,837	664	23.4%
	Yes	859	367	42.7%
YES reply to confirmation	No	3,407	902	26.5%
	Yes	289	129	44.6%
Touchpoints in 24 hours	0	2,532	571	22.6%
	1	1,044	402	38.5%
	2	119	58	48.7%
	3	1	0	0.0%

The 900 link-open events came from 859 PTPs, since some borrowers opened the link more than once. Fulfilment rises with each touchpoint from 0 to 2 (22.6%, 38.5% and 48.7%); the one PTP with three touchpoints is too few to read. Figure 14 (in Section 8) shows the index and touchpoint results side by

side. As noted in Section 8, none of these signals exists at capture time, so they are used as a second-stage check only.

9. Final feature matrix export

When all features were built, the notebook kept **25 chosen columns plus the 6 lender columns**, 31 in total, as the final feature matrix (final_features_df). The voice-bot flag (is_bot) is built in Section 5 and used in the analysis, but it is not among the exported columns. The table shows the groups, the section that explains each, and whether the model uses them.

Group	Columns	Section	Used by the model?
Identifiers and target	ptp_id, account_id, split, is_kept, annotated_type	2.3, 3	No (is_kept becomes the target)
Label reliability	confidence_weight, is_unreliable_label	3.3	No (weights not passed to training)
Agent and channel	language_team_encoded, shift_bunching_flag	5, 5.1	Yes
Language	has_hedge, has_condition, has_vague, has_firm, category_A_uncertainty, category_B_conditional, borrower_proposed_date, agent_proposed_date, seconds_to_hangup	6, 6.3	Yes (seconds_to_hangup as a number, the rest as flags)
Financial history	historical_keep_rate, is_high_risk_repeat_promiser, salary_proximity_score, hardship_score, composite_hardship_index	7, 7.1	Yes
Post-call activity	digital_touchpoints_24h, digital_intent_index_24h	8, 8.1	No (second-stage check)
Lender controls	lender_Fintech NBFC, lender_NBFC, lender_NBFC (MSME), lender_NBFC-MFI, lender_Private bank, lender_Small finance bank	3.3	Yes

The model therefore receives 5 numeric features, 11 flags and encodings, 6 lender columns and the transcript text (Section 10.4), which matches the setup described in Section 10.1.

10. Modelling

10.1 Setup

- The target is is_fake = 1 − is_kept, because the aim is to catch fake promises.
- Features: 5 numeric (hang-up time, salary proximity, historical keep rate, hardship score and index), 11 flags and encodings, 6 lender-type columns, and TF-IDF on the full transcript (words and word pairs, 1,000 features).
- Everything is in one scikit-learn pipeline (median imputation and scaling for numbers, most-frequent imputation for flags, TF-IDF for text), so each cross-validation fold learns its own settings and nothing leaks.
- Hyperparameters were tuned with 3-fold stratified GridSearchCV. The decision threshold was then chosen on the validation set, and the test set was used once.

10.2 Why these metrics

- **PR-AUC** (average precision) was the search score. It looks at precision and recall across all thresholds and is not inflated by the many true negatives that make accuracy and ROC-AUC look good.

- **F1** (2PR / (P + R)) picks the threshold, balancing fakes caught (recall) against genuine promises wrongly flagged (precision). The default 0.5 is arbitrary on imbalanced data.
- Logistic regression, decision tree and random forest used balanced class weights. XGBoost used scale_pos_weight.

10.3 Results on the test set

Model	PR-AUC	F1	Precision	Recall
Logistic Regression	0.8262	0.8421	0.7299	0.9950
Decision Tree	0.7685	0.8332	0.7140	1.0000
Random Forest	0.8278	0.8426	0.7361	0.9851
Naive Bayes	0.8231	0.8363	0.7199	0.9975
Gradient Boosting	0.8336	0.8447	0.7379	0.9876
XGBoost (final)	0.8341	0.8458	0.7425	0.9826

XGBoost scored highest and uses a shallow, slowly learning setup (depth 3, learning rate 0.03, 200 trees), which keeps variance low. The full pipeline was saved with joblib so production applies the same transformations.

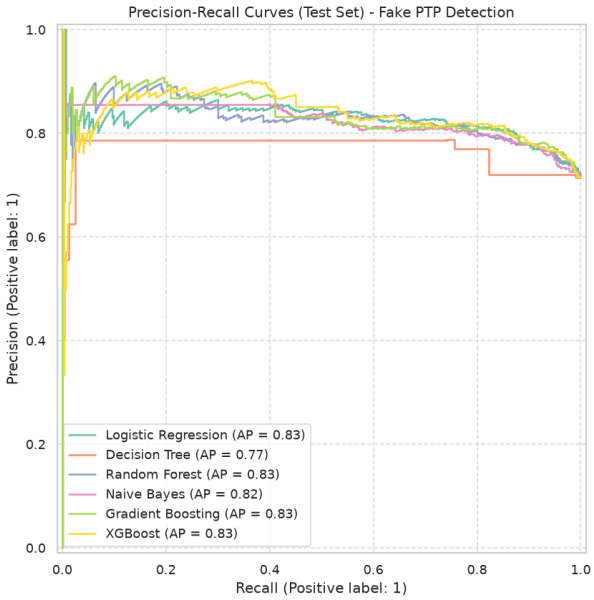


Figure 15. Precision-recall curves, test set

10.4 End-to-End Machine Learning Pipeline: detailed walkthrough

Libraries. scikit-learn provides the preprocessing, pipelines, tuning and metrics, XGBoost provides the final model, and joblib saves the finished pipeline.

Text features. For each call the turns are joined into one text with each turn prefixed by its speaker, the call is linked to its PTP through the dial attempts, and one call is kept per PTP. PTPs with no transcript get an empty text. This text is added to the feature matrix as full_text and later turned into TF-IDF features, which capture wording that the hand-made word lists miss.

Target and splits. The target is is_fake = 1 − is_kept. Rows are assigned by the pre-defined split column: train for fitting, validation for choosing the threshold, and test for the single final evaluation.

Preprocessing. A single ColumnTransformer treats each feature group in its own way, and anything not listed is dropped.

Feature group	Columns	Treatment
Numeric (5)	seconds_to_hangup, salary_proximity_score, historical_keep_rate, hardship_score, composite_hardship_index	Median imputation, then standard scaling
Flags and encodings (17)	11 flags and encodings plus 6 lender columns	Most-frequent imputation
Text (1)	full_text	TF-IDF, up to 1,000 terms, single words and word pairs

Training routine. The same function trains every model:

- 3-fold stratified cross-validation (shuffled, seed 42) searches the settings, scored by average precision (PR-AUC).
- The best pipeline predicts probabilities on the validation set, and the threshold is the point on the precision-recall curve with the highest F1.
- That threshold is applied once to the test set, where PR-AUC, F1, precision and recall are reported.
- Naive Bayes needs dense input, so a sparse-to-dense step is added to its pipeline.

Settings searched and chosen. Together the six searches test 34 setting combinations, each in 3 folds.

Model	Settings searched	Combos	Chosen settings
Logistic Regression	C = 0.1, 1.0, 10.0 (balanced class weights)	3	C = 1.0
Decision Tree	max depth 3, 5, none; min samples to split 2, 10 (balanced)	6	Depth 3, min split 2
Random Forest	Trees 100, 200; max depth 5, 10, none (balanced)	6	200 trees, depth 10
Naive Bayes	var_smoothing 1e-9, 1e-8, 1e-7	3	1e-7
Gradient Boosting	Trees 100, 200; max depth 3, 4	4	100 trees, depth 3
XGBoost	Trees 200, 400; max depth 3, 4, 5; learning rate 0.03, 0.05; subsample 0.8; column sample 0.8	12	200 trees, depth 3, learning rate 0.03, subsample 0.8, column sample 0.8

XGBoost also uses scale_pos_weight, set to the number of non-fake PTPs divided by fake PTPs in the training split, and tracks the area under the precision-recall curve while it trains. The test results for all six models are in Section 10.3, and Figure 15 plots their precision-recall curves.

Saving the model. The complete XGBoost pipeline, including the preprocessor, the TF-IDF vectoriser and the tuned model, is saved with joblib as fake_ptp_xgboost_pipeline.pkl. The decision threshold chosen on the validation set is not stored inside this file, so it should be recorded and applied separately when the model is used.

11. How the analysis shaped the solution

Finding	What we did
Payments agree with human labels (V = 0.56)	Used the payment-based label as the target
Ring time, propensity, portfolio, lender, channel not significant	Dropped raw versions; kept lender columns as cheap controls
Amount columns collinear; paid_amount leaks the target	Excluded them; used ratio-style features
Late-shift surge; small language-team differences	Shift bunching flag; language-team encoding
Tenure not significant	Not used
Firm +22.8 pp, vague −7.1 pp, uncertainty 17.5% kept	Four language flags, two category flags, TF-IDF
Borrower-date idea reversed; hang-up time weak	Kept as weak features
Keep-rate groups; salary distance	Historical keep rate, high-risk flag, salary proximity
Post-call intent 14.6% to 53.9%	Held back as a second-stage check

12. Difficulties, assumptions and limitations

Difficulties we worked through

- Mixed date formats and repeated column names after merging.
- Code-mixed and partly Devanagari transcripts.
- A very small labelled sample, handled with rank-based tests, effect sizes and Wilson intervals.
- Sparse fields: salary day exists for 1,162 PTPs and hang-up time for 3,067 of 3,696.
- Avoiding leakage: train-only fitting, preprocessing inside the pipeline, and one use of the test set. Naive Bayes needed dense input, so a sparse-to-dense step was added.

Assumptions

- The 80% payment rule, 3-day grace window and EMI fallback; payments are matched to a PTP by account and time window only.
- Shift end of 19:00 and a 30-minute window, both inferred from the data.
- Word lists are hand-made from phrases we saw; other wording is only covered by TF-IDF.

Limitations we noticed on review

- About 72% of PTPs are fake, so fake is the majority class, not the minority. The Decision Tree's precision of 0.714 at recall 1.0 implies a test base rate near 0.71. A rule that flags everything would score F1 of about 0.83, so our F1 of 0.846 is only a small step above it. PR-AUC of 0.834 against a 0.71 base rate shows real but moderate ranking ability.
- Picking the threshold by maximum F1 pushes recall close to 1. A better choice would use business cost, such as precision at a fixed recall, and should report lift over the base rate. scale_pos_weight is also below 1 here, which down-weights fakes.
- Language-team encoding is an unsmoothed mean, confidence weights were not passed to training, and the top models differ by less than 0.01 in PR-AUC, so the choice between them is close.

13. Conclusion

We built a payment-based label, confirmed it against human judgement, and tested each idea before using it. The most useful capture-time signals were firm commitment language, vague timing, uncertainty

phrases, repeat-promiser history and distance from payday. Post-call digital activity was the strongest signal overall and is best used as a second-stage check. The final tuned XGBoost pipeline (PR-AUC 0.834, F1 0.846) is our submitted model. The clearest next improvement is choosing the threshold by business cost instead of maximum F1.